

Direction 2: atmost one of $P_1, P_2, \dots, P_n$ is true $\Rightarrow \bigwedge_{i=1}^{n-1} \bigwedge_{j=i+1}^n (\neg P_i \vee \neg P_j)$ is True.

Proof: **Case 1:** $\forall i \in N, 1 \leq i \leq n, P_i$ is False.

$\therefore$ By defn. of negation, $\neg P_i$ is True.

By defn. of OR, $\forall i \in N, 1 \leq i \leq n, (\neg P_i \vee \neg P_j)$ is True.

By defn. of AND, $\bigwedge_{j=i+1}^n (\neg P_i \vee \neg P_j)$ is True.

By defn. of AND, $\left[\bigwedge_{i=1}^{n-1} \bigwedge_{j=i+1}^n (\neg P_i \vee \neg P_j) \right]$ is True.

Case 2: ~~Let P_i be True. The rest be False.~~

$\therefore$ By defn. of negation, $\neg P_i$ is False. But except P_i , $\forall i \in N, 1 \leq i \leq n$, $\neg P_j$ must be True.

By defn. of OR, $(\neg P_i \vee \neg P_j)$ is True. By defn. of AND, $\left[\bigwedge_{j=i+1}^n (\neg P_i \vee \neg P_j) \right]$ must be True.

By defn. of AND, $(\neg P_i \vee \neg P_k)$, where $j \neq i$ and $k \neq i$ must be True.

$\therefore \left[\bigwedge_{i=1}^{n-1} \bigwedge_{j=i+1}^n (\neg P_i \vee \neg P_j) \right]$ must be True by defn. of AND.

45) Use Exercise 44 to construct a compound proposition that is true if and only if ~~one~~ exactly one of the propositions $P_1, P_2, \dots, P_n$ is true.

Soln: **Claim:** $\left[\bigwedge_{i=1}^{n-1} \bigwedge_{j=i+1}^n (\neg P_i \vee \neg P_j) \right] \wedge (P_1 \vee P_2 \vee P_3 \vee \dots \vee P_n)$ is true if and only if exactly one of the propositions $P_1, P_2, \dots, P_n$ is true.

Proof: **Direction 1:** $\left[\bigwedge_{i=1}^{n-1} \bigwedge_{j=i+1}^n (\neg P_i \vee \neg P_j) \right] \wedge (P_1 \vee P_2 \vee \dots \vee P_n)$ is true $\Rightarrow$ exactly one of the propositions $P_1, P_2, \dots, P_n$ is true.

~~By Defn. of AND, $\left[\bigwedge_{i=1}^{n-1} \bigwedge_{j=i+1}^n (\neg P_i \vee \neg P_j) \right]$ and $(P_1 \vee P_2 \vee \dots \vee P_n)$ are respectively true.~~